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A

Power – 12V, solar and USB inputs, 18650 charger, rails

File: power.kicad_sch

ESP32, programming, buttons, RTC, LED strip, buzzer

File: mcu.kicad_sch

RS-485 and uplink

File: comms.kicad_sch

B

AgriNode HUB-C (4G hub)

One board per hub type: HUB-W (Wi-Fi), HUB-L (LoRa), HUB-C (4G LTE module board), GW-LAN (LoRa to Ethernet)

Core: ESP32-WROOM-32E, 12 V / solar / USB-B inputs, 18650 + TP4056 + power path, MIC29302 3.3 V, DS3231M RTC

USB programming (CH340C, auto-reset) + PROG header, RESET and BOOT/PAIR, WS2813 LED-strip header, buzzer

Optional RS-485 (Modbus) – leave unpopulated if not needed

Source of truth: design/hubs.py – regenerate with build.py <variant>

C

D

E

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F

Generated by tools/kigen from the Python design source – edit the source, not this file

AgriNode

Sheet: /

File: agrinode_hub_c.kicad_sch

Title: AgriNode HUB-C (4G hub) – Overview

Size: A3

Date: 2026-09-27

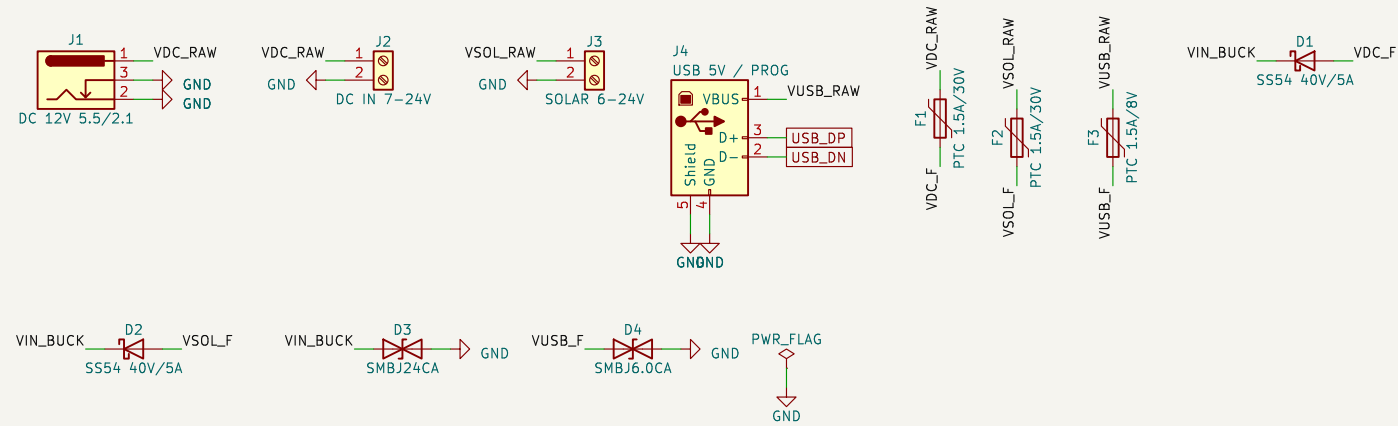
Rev: A

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

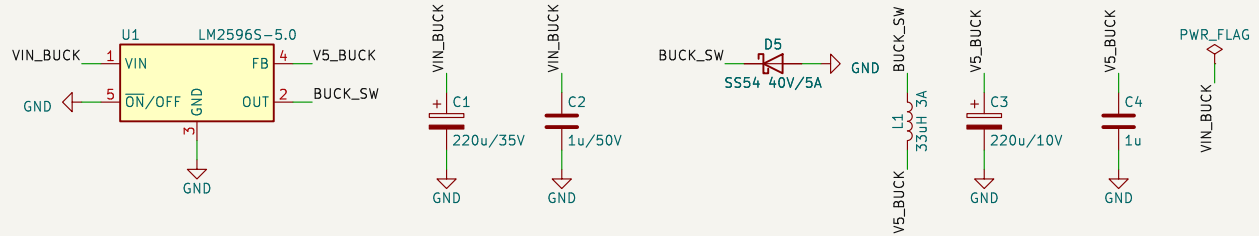
Id: 1/4

Inputs: 12 V adapter (jack + terminal), solar 6–24 V, USB 5 V

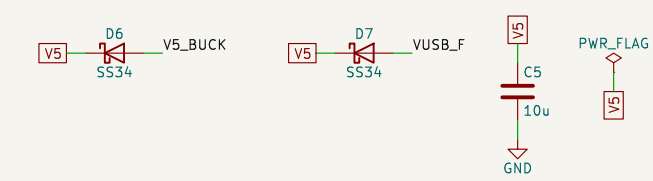
each input fused and reverse–polarity protected; any combination may be connected



5 V buck (LM2596S–5, 7–40 V in, 3 A)

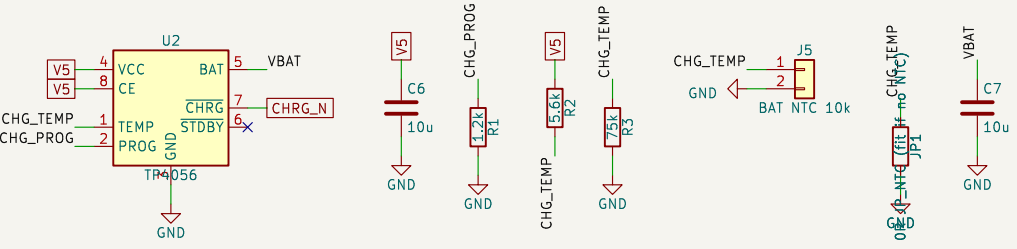


5 V bus (diode–OR of buck and USB)

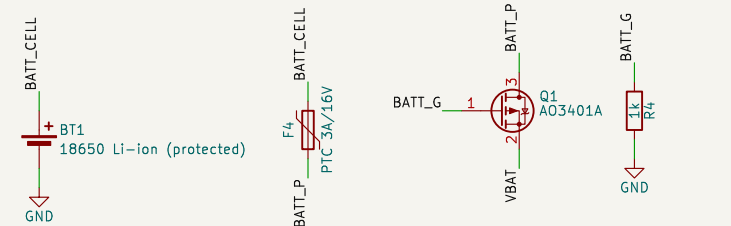


18650 charger (TP4056, 1 A) with battery temperature guard

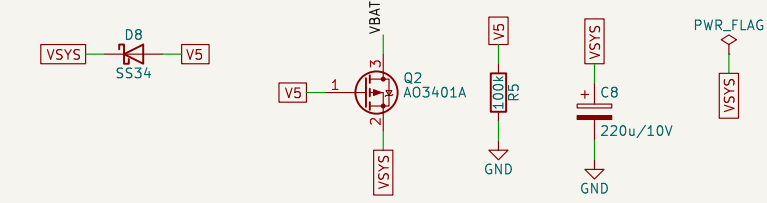
NTC 10k B3950 on the cell; charge only 0..45 C. No NTC: fit JP_NTC (TEMP to GND)



18650 cell: holder, fuse, reverse–insertion protection

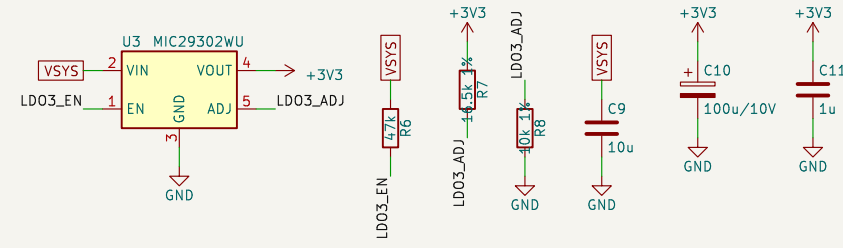


Power path: 5 V bus when present, battery otherwise (no gap)



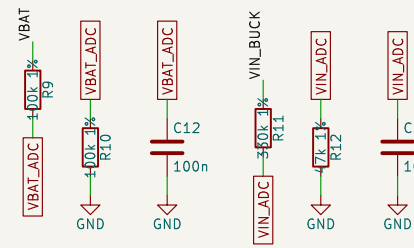
3.3 V rail (MIC29302, low dropout, 3 A)

$V_{out} = 1.24 \times (1 + 16.5k/10k) = 3.29 \text{ V}$

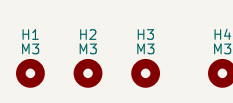


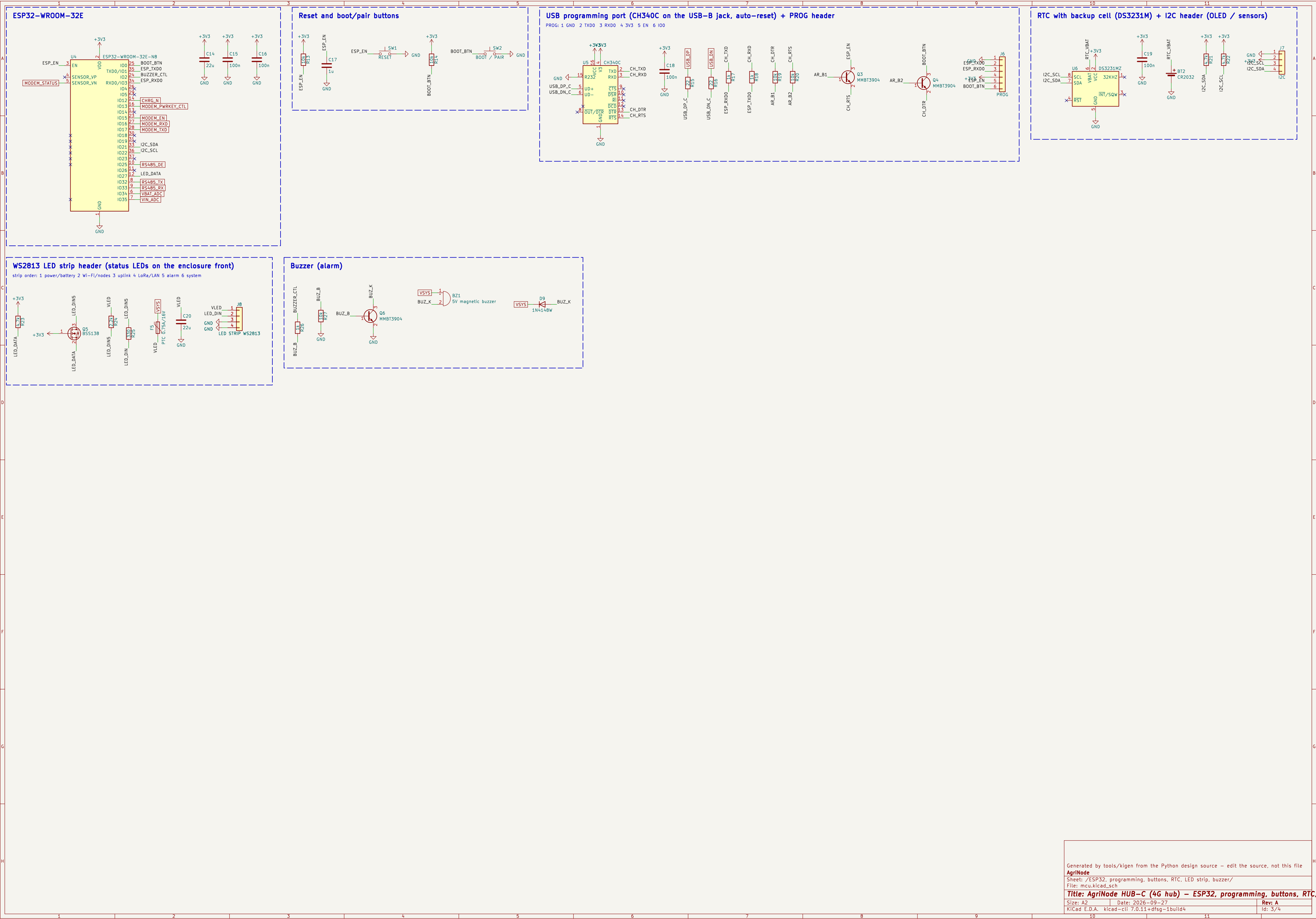
Supply monitoring (ADC1)

VBAT: 100k/100k VIN: 330k/47k (24 V -> 3.0 V)

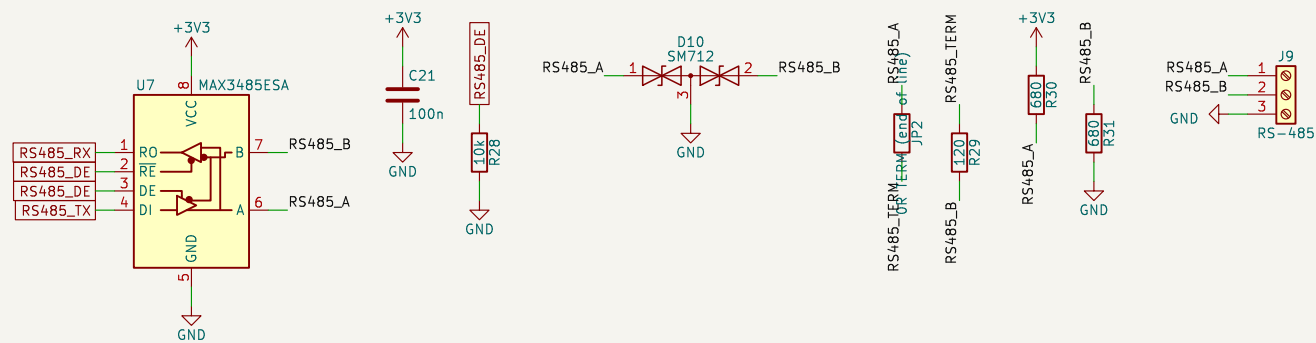


Mounting holes (M3)





RS-485 / Modbus RTU (wired industrial sensors, up to 1 km)



4G LTE carrier-board slot (SIMCom A7670E / Quectel EC200U module board)

switched supply for power-cycling the modem: 3.3 V UART

